

PREETI GURAPU

Sr. Cloud Engineer

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Professional Summary:

Accumulated 10+ years of professional IT experience encompassing various domains such as Software Development, DevOps, Testing, Configuration Management, Build and Deployment Automation, Source Code Repository Management, and Environment Management.

- Proficiently managed application software implementation and maintenance across diverse Web and Cloud/Server Environments, including Amazon Web Service (AWS) and Azure.
- Demonstrated expertise in DevOps practices, Release Engineering, Configuration Management, and Cloud Infrastructure Automation tools. Proficient with Git, Bitbucket, Maven, Jenkins, TeamCity, Chef, Ansible, Terraform, Docker, Kubernetes and ELK.
- Engaged in the entire lifecycle of services, from inception and design to deployment, operation, and refinement.
- Embedded DevSecOps practices into CI/CD pipelines, improving early detection of security vulnerabilities.
- Leveraged Terraform to automate the provisioning of VPCs, ELBs, Security Groups, SQS queues, and S3 buckets, transitioning towards infrastructure as code (IaC).
- Designed and managed Kubernetes clusters on cloud platforms such as AWS EKS, Azure AKS, and Google GKE, ensuring high availability, scalability, and fault tolerance.
- Utilized programming skills in Java, Python, Go, Perl, Ruby, C, or C++ to automate and troubleshoot distributed systems.
- Developed Python scripts to automate repetitive tasks such as log monitoring, data extraction, and report generation.
- Extensive experience in provisioning IAAS and PAAS concepts in cloud computing platforms such as AWS, Azure, and Google Cloud Platform, while adhering to Agile, Scrum, and Waterfall methodologies.
- Implemented DevOps practices for Micro Services and Docker-based clusters, employing Kubernetes and Spinnaker as orchestrators for cloud platform application development, with a focus on Spring Boot and Microservices.
- Built & maintained Docker container clusters, managed AWS environments, & utilized technologies like Linux, Bash, GIT& Docker. Employed Kubernetes pods & Docker to create a runtime environment for the CI/CD system, facilitating efficient build, testing, & deployment.
- Implemented advanced Kubernetes features, including Horizontal Pod Autoscaling (HPA), Cluster Autoscaler, and Custom Resource Definitions (CRDs) for tailored application needs.
- Designed and developed enterprise-grade applications using Java and JavaScript, ensuring high performance and scalability.
- Conducted threat analysis to identify potential risks and implement mitigating strategies during development phases.
- Proficiently configured CI (Continuous Integration) pipelines, with expertise in Jenkins and TeamCity server setup, script creation (Bash shell, Perl, Python), and VM configuration (Windows/Linux). Additionally, generated Infrastructure reports using PowerShell scripts.
- Extensive experience in troubleshooting, configuring, and deploying enterprise applications on WebSphere, WebLogic, Tomcat, and JBoss, while conducting SCM verifications and audits to ensure SOX compliance.
- Proficient in addressing database performance-related issues in MySQL and Oracle databases.
- Demonstrated hands-on expertise with configuration management tools such as Chef, Puppet, and Ansible, along with a strong commitment to continuous learning and tool adoption.
- Conducted performance analysis and optimization of Kubernetes clusters to enhance scalability and resource utilization.
- Optimized Kubernetes cluster performance by configuring resource quotas, limits, and node selectors, enabling efficient workload scheduling and cost-effective resource usage.
- Capable of efficiently handling multiple tasks, working both collaboratively in a team and independently to meet project deadlines.
- Partnered with teams to enforce secure coding standards and address application vulnerabilities.

- Conducted vulnerability assessments and implemented secure coding fixes for critical application modules.
- Collaborated with the security team to address findings from application scanning tools and minimize risks.
- Automated build, test, and deployment workflows using Jenkins to streamline CI/CD pipelines.
- Proficiently submitted implementation requests to various stakeholder groups involved in the release process.
- Strong background in system administration, including installation, upgrading, patch management, migration, configuration, troubleshooting, security, backup, and disaster recovery in environments spanning AWS, Azure, AKS, and OpenStack.
- Utilized OpenShift for container orchestration alongside Kubernetes, focusing on container storage, automation, and multi-tenancy enhancement within the container platform.
- Applied hands-on experience in designing modules, creating playbooks, and configuring inventories in Ansible.
- Expertise in server monitoring, capacity planning, and application monitoring, employing tools such as Splunk, GitHub/GitLab, AWS CloudWatch, and Kibana.
- Developed and implemented robust automation scripts using Python and Bash to streamline Kubernetes operations, reduce manual tasks, and enhance operational efficiency. Leveraged tools like Ansible and Terraform for infrastructure as code (IaC).
- Diagnosed and resolved production issues in Kubernetes clusters, such as node failures, pod eviction, and network bottlenecks, optimizing mean time to recovery (MTTR) and ensuring system reliability.
- Monitored and mitigated cloud-based security vulnerabilities using automated scanning tools.
- Managed different infrastructure resources, like physical machines, VMs and even Docker containers using Terraform.
- Worked with Ansible (automation tool) to automate the process of deploying/testing the new build in each environment, setting up a new node and configuring machines/servers.
- Written Cloud formation templates and deployed AWS resources using it.
- Excellent written and verbal communication skills, strong organizational skills, and a hard-working team player.

Technical Skills:

Cloud Platforms	Microsoft Azure, Aws Cloud, Google Cloud Platform
AWS Services	RDS, EC2, VPC, IAM, Lambda Function, Cloud Formation, EBS, S3, ELB, Auto Scaling, Cloud Trail, SQS, SNS, SWF, Cloud Watch, WAF, ECR, EKS.
Azure Services	App Services, Key vault, function app, Blob storage, Azure Active Directory (Azure AD), Service Bus, Azure Container Registry (ACR) and Azure Kubernetes service (AKS), Azure SQL, Azure Cosmos DB.
CI/CD	Jenkins, Azure DevOps Pipelines, CircleCI, GitLab.
Version Control Tools	GIT, Bit Bucket.
Configuration & Automation Tools	Ansible, Puppet, Chef
Container Platforms	Docker, Kubernetes, Red Hat Open Shift.
Monitoring Tools	Nagios, Splunk, Prometheus, Grafana, Datadog, New Relic.
Artifactory	Jfrog and Nexus
Web Servers	Apache Tomcat, WebSphere, Nginx
Documentation	Confluence
Operating Systems	Microsoft Windows XP/ 2000, Linux, UNIX.
Tracking Tools	Jira, ServiceNow, Azure Board.
Code Scanning	Sonar Qube, Jfrog X ray, ECR Inspector
Databases	RDS, Cosmos DB, My SQL, Aurora DB, Mongo DB.
Logging	Cloud Watch, Cloud Trail, Azure App Insights, Azure Monitor, ELK, Fluent
Languages	Python, Shell scripting, Golang.

Certifications:

- Certified Kubernetes Administrator.
- Certified AWS Developer Associate.
- Certified Azure Administrator Associate.

PROJECT EXPERIENCE:

Client	PNC – Atlanta, GA.	March 2023 to present
Role	Sr. Cloud Engineer	
Responsibilities	▪ Designed and administered AWS S3 buckets, implementing lifecycle policies and leveraging S3 Glacier for secure, cost-effective storage and backup solutions within the AWS ecosystem. ▪ Leveraged AWS Identity and Access Management (IAM) to define fine-grained permissions and manage user roles, enabling secure access to AWS resources. ▪ Architected and automated server builds, monitoring, and deployments using Ansible, improving consistency and efficiency across infrastructure. ▪ Established proactive monitoring and alerting mechanisms via AWS CloudWatch and Prometheus, tracking performance metrics like CPU utilization and disk usage, and implementing optimizations. ▪ Designed and maintained Kubernetes clusters, including pods, services, deployments, stateful sets, and persistent storage volumes, ensuring optimal performance, scalability, and availability. ▪ Extensive experience configuring and managing cloud-based Kubernetes clusters on platforms like AWS EKS, Azure AKS, and Google GKE, implementing resource provisioning, configuration management, and security best practices. ▪ Developed automation scripts in Python and Bash to streamline Kubernetes operations, including creating Helm charts for efficient deployment and management of containerized applications. ▪ Configured robust AWS network architectures, including VPCs, subnets, route tables, Internet gateways, NAT gateways, and security groups, optimizing network security and performance. ▪ Deployed containerized applications using Docker and orchestrated them with Kubernetes, ensuring high availability, load balancing, and horizontal scaling of microservices. ▪ Designed CI/CD pipelines using Jenkins, GitLab CI, and AWS CodePipeline, incorporating automated builds, testing, security scans, and deployments to Kubernetes clusters. ▪ Implemented Terraform and CloudFormation for Infrastructure as Code (IaC), provisioning EC2 instances, ELBs, S3 buckets, and Kubernetes clusters with consistency and reliability. ▪ Configured AWS Multi-Factor Authentication (MFA) to enforce enhanced security for user accounts, utilizing tools like AWS Virtual MFA and Google Authenticator. ▪ Deployed and managed logging and monitoring solutions using ELK Stack (Elasticsearch, Logstash, Kibana), Splunk, and Fluentd, centralizing log data from Kubernetes clusters and AWS resources. ▪ Enhanced Kubernetes cluster security by implementing network policies, role-based access control (RBAC), and automated Pod Security Policies (PSP) for compliance with best practices. ▪ Built and deployed stateful applications on Kubernetes using StatefulSets with persistent volumes, ensuring data durability and high availability. ▪ Orchestrated workflows with Apache Airflow and Kubernetes CronJobs to schedule ETL jobs and automate recurring tasks in a containerized environment. ▪ Integrated advanced logging and metrics monitoring using Prometheus, Grafana, and AWS CloudWatch, enabling detailed insights into Kubernetes cluster health and application performance. ▪ Diagnosed and resolved complex Kubernetes issues, including pod scaling, storage binding, and network connectivity, ensuring minimal downtime and efficient resource utilization. ▪ Worked with tools like K9s, Kubectrl, and Lens to manage Kubernetes clusters effectively and debug live environments. ▪ Designed and implemented serverless functions using AWS Lambda for event-driven data processing and integrated them with containerized backend services. ▪ Applied OWASP Secure Coding Practices and built security measures into containerized applications, ensuring compliance with industry standards.	

	- Developed and maintained automation scripts for provisioning Kubernetes clusters and deploying microservices using Helm and Kustomize, reducing deployment time and errors. - Collaborated with cross-functional teams to integrate Java-based back-end APIs with modern React.js front-end frameworks, ensuring a seamless user experience. - Configured DNS infrastructure using Amazon Route 53, managing DNS zones and associating public DNS names with Kubernetes ingress controllers. - Conducted in-depth performance tuning for Kubernetes clusters by optimizing resource quotas, node affinity, taints, and tolerations, ensuring efficient workload distribution.
Environment	AWS, Terraform, Chef, Ansible, YAML, Docker, Kubernetes, Openshift, EKS, Azure CLI, Jenkins, AKS, Git, Maven, Splunk, Nagios, Python Scripts, Kickstart, Java/J2EE, Linux

Client	OPTUM - Eden Prairie, MN.	Dec 2021 to Feb 2023
Role	Sr. Cloud Engineer	
<u>Responsibilities</u>	- Orchestrated Slot Blue-Green Deployments for Azure App Services. - Containerized Jenkins pipelines during the migration to Azure DevOps. - Leveraged Azure services, including Azure DevOps, Azure Kubernetes Services (AKS), Container Registry, Cosmos DB, Grafana, Azure Pipelines, Monitor, and RBAC for microservices development, application migration, and management. - Deployed Azure IaaS virtual machines (VMs) and Cloud services (PaaS role instances) into secure VNets and subnets. - Configured cloud-based security solutions such as RASP, WAF, and vulnerability scanners to safeguard applications. - Developed solutions with UNIX/Linux systems and scripting using Bash, Shell, or similar tools. - Performed static and dynamic application security testing using tools like Fortify, Checkmarx, and Veracode. - Implemented role-based access control (RBAC) and Pod Security Policies (PSP) in Kubernetes clusters, ensuring strict adherence to security and compliance standards. - Written Templates for Azure infrastructure as a code using Terraform to build staging and production environments. Integrated Azure Log Analytics with Azure Virtual machines for monitoring the log files, store them and track metrics - Implemented Ansible as a configuration management system for web applications, environment configurations, users, mount points, and package management. - Engineered pre-commit hooks for GitHub/GitLab integration with Amazon EKS, Jenkins, and JIRA, complemented by post-commit hooks for automated continuous builds. - Certified Kubernetes Administrator (CKA), Certified Kubernetes Security Specialist (CKS), and Certified Kubernetes Application Developer (CKAD). - Automated build, test & deployment processes by tightly integrating Git with CI/CD pipelines. - Orchestrated middleware artifact deployments through Jenkins job configurations. - Configured intricate cloud infrastructures, featuring Virtual Networks, Virtual Machines, Azure Active Directory, Load Balancers, Azure SQL, and Cloud Services. - Implemented cloud-based CI/CD pipelines, leveraging services like AWS CodePipeline and Azure DevOps. - Architected, deployed, and fine-tuned hybrid cloud components such as Azure AD, ADFS, SSO, and VPN Gateways to ensure uninterrupted business operations. - Spearheaded the development and maintenance of Splunk-based monitoring and alerting systems, resulting in a 40% reduction in mean time to resolution (MTTR) and a 25% increase in system availability. - Automated application health monitoring and alerting mechanisms using Python-based scripts. - Employed advanced Splunk configurations to filter and manage log data effectively, including the utilization of Splunk as a SIEM solution for real-time security monitoring.	

	- Managed and maintained multiple Web Logic Servers, guaranteeing load balancing, high availability, and seamless failover capabilities. - Excelled in Docker containerization, crafting Docker files, orchestrating automated builds, and streamlining multi-container provisioning with Docker Compose. Further streamlined container management via Azure Container Registry. - Pioneered cluster management solutions using Docker and AKS to facilitate local Kubernetes deployments, including the creation of self-hosted Kubernetes clusters through Jenkins & Git. - Showcased Kubernetes proficiency through the creation of pods, controllers, services, deployments, health checks, and ingress configurations using YAML files. - Utilized Terraform templates for Azure infrastructure provisioning, featuring virtual machines, scale sets, Azure Site Recovery, and Azure Backup. Further enhanced infrastructure control through Log Analytics data collection. - Integrated monitoring and logging tools - Prometheus, Grafana, Fluentd, and ELK Stack to provide actionable insights into Kubernetes cluster and application performance. - Demonstrated expertise in Ansible, creating modular components, composing intricate playbooks, and meticulously configuring inventory. - Spearheaded the implementation of an ELK stack for centralized logging, driving real-time log analysis, anomaly detection & swift issue resolution, culminating in a 30% reduction in MTTR. - Proficiently managed Private Cloud and Hybrid Cloud configurations, spanning Windows Azure, SQL Azure, Azure Web, and Database deployments. - Developed PowerShell scripts for orchestration and harnessed Dynatrace for monitoring and support.
Environment	Azure, Terraform, Chef, Ansible, Docker, Kubernetes, Jenkins, Git, Maven, Splunk, Nagios, Python Scripts, Kickstart, Java/J2EE, Linux.

Client	State of NE – Lincoln, NE	Jun 2020 to Nov 2021
Role	Sr. Cloud Engineer	
<u>Responsibilities</u>	- Worked on many different concepts of Chef like Roles, Environments, Data Bags, Knife, and Chef Server Admin/Organizations. - Written Chef Recipes to automate build/deployment process and data bags in Chef for better environment management. - Tested Chef Cookbook modifications on cloud instances in AWS and using Test Kitchen and Chef Spec - Written Chef Cookbooks for various DB configurations to modularize and optimize product configuration, converting production support scripts to Chef Recipes and AWS server provisioning using Chef Recipes. - Designed, implemented, and maintained Kubernetes clusters, ensuring optimal resource provisioning, configuration management, and adherence to security best practices. Managed both on-prem and cloud-based Kubernetes environments. - Used SonarQube for continuous inspection of code quality and to perform automatic reviews of code to detect bugs. - Automated Nagios alerts and email notifications using Python script and executed them through Chef. - Installed, configured and maintained web servers like Apache Web Server and WebSphere Application Server on Red Hat Linux. - Experience with Red Hat Linux kernel, memory upgrades and swaps area. - Red Hat Linux Kickstart. Installation Sun Solaris Jumpstart Installation. Configuring DNS, DHCP, NIS, NFS in Sun Solaris 8/9 & other Network Services. - Used multiple EC2 instances at the same time and provided highly durable and available data by using S3 data store, versioning, lifecycle policies, & create AMIs for mission-critical production server's backup.	

	▪ Provided highly durable and available data by using S3 data store, versioning, lifecycle policies. ▪ Performed the automation deployments using AWS by creating the IAMs and used the code pipeline plugin to integrate Jenkins with AWS and created the EC2 instances to provide the virtual servers. ▪ Worked with Knife command line tool for creating Recipes, Cookbooks and worked with chef supermarket. ▪ Implemented Docker-Maven plugin and Maven POM to build Docker Images for all microservices and later used Docker file to build the Docker Images from the java jar files. ▪ Used Git for source code version control and integrated with Jenkins for CI/CD pipeline, code quality tracking and user management with build tools Maven and Ant and written Maven pom.xml build script. ▪ Worked on NoSQL database MongoDB to replica setup and sharing. ▪ Also experienced in managing replica set. ▪ Installed, Configured, and Managed Monitoring Tools such as Nagios for Resource Monitoring/ Network Monitoring.
Environment	AWS (EC Route, EBS, Security Group, Auto Scaling, and RDS), GIT, Chef, Docker, SonarQube, Maven, Jenkins, ANT, Python, Nagios.

Client	Ally Financial - Charlotte, NC.		Jul 2017 to May 2020
Role	Sr. Cloud Engineer		
<u>Responsibilities</u>	▪ Experience in creating Docker containers leveraging existing Linux containers and AMIs in addition to creating Docker containers from scratch. ▪ Worked on creation of custom Docker container images, pushing the images and Docker consoles for managing the application lifecycles. ▪ Utilized key features in Terraform such as Infrastructure as Code (IAC), Change automation, Execution plans. Wrote Terraform templates to deploy infrastructure on cloud with EC2, ELB. ▪ Configured AWS IAM and security groups in Public and Private subnets in VPC. ▪ Worked on designing and deploying applications using AWS services like EC2, Route53, S3, RDS, DynamoDB, SNS, SQS and ELK. Focused on high-availability, fault tolerance and auto scaling in AWS Cloud Formation. ▪ Involved in building configuration management for Microservices using Docker, Kubernetes and Ansible. Used Kubernetes to setup load balancing and service selection. ▪ Worked on managing Private cloud environment using Ansible and Enhanced the automation to assist configuration management using Ansible based YAML scripts. ▪ Worked on Docker container snapshots, attaching to a running container, removing images, managing Directory structures, and managing containers. ▪ Managed local deployments in Kubernetes, created local cluster and deploying application containers. created private cloud using Kubernetes that supports Dev, Test and Prod environments. ▪ Configured Jenkins for doing the build in all the non-production and production environments. ▪ Used Git with Jenkins to integrate to automate the code checkout process. ▪ Configured Jenkins for doing the build in all non-production environments and most of the production environments.		
Environment	AWS, Docker, Kubernetes, Jenkins, YAML, Ansible, Terraform, GIT, MAVEN, Jenkins, AWS Cloud, Docker, PowerShell.		

Client	Aveva, Hyderabad, India.	Jun 2013 to Feb 2016
Role	Linux Administrator	
<u>Responsibilities</u>	▪ Installation, configuration, backup, recovery, maintenance and support of RedHat Linux and Solaris. ▪ Troubleshooting Day-to-Day issues with various Servers on different platforms. ▪ File system Administration, Setting up Disk Quota, solutions on SAN/NAS storage. ▪ Configuration and administration of Clustered servers on SUSE Linux environment. ▪ Configuring backups on newly built servers and monitoring failed backups. ▪ Install and configure Ubuntu, Centos on remote servers and Desktop servers. ▪ Installation of patches and packages using RPM and YUM in Red hat Linux. ▪ Installed and configured Apache/Tomcat web server. ▪ Provided 24x7 on-call supports in debugging and fixing issues related to Linux, Solaris. ▪ HP-UX Installation/Maintenance of Hardware/Software in Production, Development. ▪ Responsible for maintenance of source control systems Subversion and GIT. ▪ Created repositories according to the structure required with branches, tags, and trunks. ▪ Created hooks and managed permissions on the branches for GIT. ▪ Responsible for designing and deploying best SCM processes and procedures. ▪ Installed and configured SSH server on Red hat/Cent OS Linux environments. ▪ Created and maintained several build definitions and publish profiles to handle automated builds in an Agile/SCRUM environment. ▪ Attended sprint planning sessions and daily sprint stand-up meetings. Configured application servers (JBOSS) to deploy the code. ▪ Used Apache Tomcat as application server for deploying the artifacts. ▪ Installed, administered, and configured Jenkins/Bamboo Continuous Integration tool. ▪ Used Ant, Maven to perform daily and weekly Software Builds. ▪ Helped developers and other project teams to set views and environments. ▪ Replicated the Jenkins build server to a test VM using Packer, Virtual Box, Vagrant, Chef, Perl brew and Server spec.	
Environment	Linux, UNIX, RHEL, CentOS, Ubuntu, NIS, DHCP, FTP, Telnet, Nagios, Kickstart, SSH, VM Sphere, VMWare Virtual Box, RPM and YUM, Logical Volume Manager (LVM), Internet Protocol Suite (TCP/IP), Tomcat Apache, Firewalls, VMware, Unix.	